

Georg von Westphalen

MedKayfab: Neurology Edition

Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kard Kult
// Klub for Narrative Medical Learning Games *DR*

Stay
fluffy!

BLÖDSINN!



The Plumber's Cousin

(TACI)

3/3
Deficits



POWER: Total Shutdown. Contralateral hemiplegia, homonymous hemianopia, AND higher cortical dysfunction (dysphasia/neglect).

LORE: High mortality, severe disability. Proximal MCA or ICA occlusion.

Manny MCA

(Middle Cerebral Artery)

Arm > Leg
Weakness



POWER: Dominant Disconnect. Contralateral hemiplegia + hemisensory loss. If dominant: Aphasia. Non-dominant: Neglect.

LORE: The most common stroke territory. Follows the motor homunculus.

Annie ACA

(Anterior Cerebral Artery)

Leg > Arm
Weakness



POWER: Frontal Filter Failure. Contralateral leg weakness + sensory loss. Urinary incontinence + personality change (abulia).

LORE: Hits medial frontal/parietal cortex. The filter is gone.

Vic Vertebral

(Posterior Circulation)

The 5 Ds



POWER: Cross-Wiring. Produces crossed signs: ipsilateral cranial nerve palsy with contralateral long tract signs.

LORE: Dizziness, Diplopia, Dysarthria, Dysphagia, Drop attacks. Highly dangerous to misdiagnose in the ED.

Larry Lacune

(Lacunar Infarct)

< 15mm
Infarct



POWER: Pure Precision. Pure motor OR pure sensory stroke. NO cortical signs (no aphasia, neglect, or hemianopia).

LORE: Caused by lipohyalinosis of deep perforators. Strongly linked to hypertension and diabetes.

Wally Wallenberg

(Lateral Medullary Syndrome)

Touch
Spared



POWER: Crossed Pain. Ipsilateral face pain/temp loss + contralateral body pain/temp loss. Ipsilateral ataxia and Horner's.

LORE: PICA occlusion. Touch and proprioception are completely spared!

Benny Basilar

(Basilar Artery Occlusion)

> 6h
Window



POWER: The Lock-In. Quadriplegia, anarthria, but consciousness and vertical gaze are fully intact.

LORE: A devastating natural history. One of the few syndromes where thrombectomy is attempted well beyond 6 hours.

Otto Occipital

(Posterior Cerebral Artery)

Macular
Sparing



POWER: Alexia Without Agraphia. Contralateral homonymous hemianopia. Can write, but cannot read (if dominant).

LORE: Macular vision is spared due to dual blood supply (MCA collaterals at the occipital pole).

Tia the Transient

(TIA)

STAT Block

ABCD2 Score

POWER

The Warning Shot

Focal ischemia resolving within 24h (most <60m).
High ABCD2 score mandates same-day review.

LORE

Requires urgent dual antiplatelet therapy for high-risk patients.
Do not ignore.



Sam the SAH

(Subarachnoid Haemorrhage)

STAT Block

< 6h CT >98%

POWER

Thunderclap

Worst headache of life, maximum severity at onset.
Due to ruptured berry aneurysm.

LORE

Ottawa SAH Rule:
If CT is negative after 6h, mandatory LP for xanthochromia.
Vasospasm risk days 4-14.



The Penumbra Princess

(Ischaemic Penumbra)

STAT Block

4.5h Window

POWER

Salvage Operation. The target of all acute stroke therapy. Brain tissue that is ischemic but not yet dead.

LORE

IV alteplase eligible within 4.5 hours.
Thrombectomy extends to 24h with favorable perfusion imaging.



Hank the Hemorrhage

(Intracerebral Haemorrhage)

STAT Block

SBP < 140 within 1h

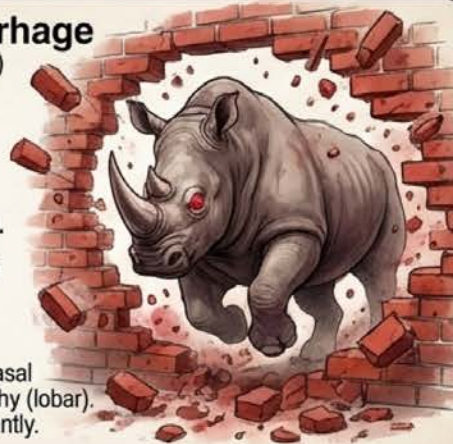
POWER

Absolute Contraindication.

Sudden focal deficit + headache, vomiting, reduced consciousness. NO tPA.

LORE

Hypertensive small vessel (basal ganglia) vs. Amyloid angiopathy (lobar).
Reverse anticoagulation instantly.



What the FLUFF?!

The NIHSS Inspector

(NIH Stroke Scale)

STAT Block
Max Score: 42

POWER

Predictive Threshold.
Score ≥ 6 suggests
large vessel occlusion.
Score < 4 is minor stroke.

LORE

A dynamic, serial scoring tool.
Essential for predicting the
need for thrombectomy.



Cissy Circle of Willis

(Anatomy)

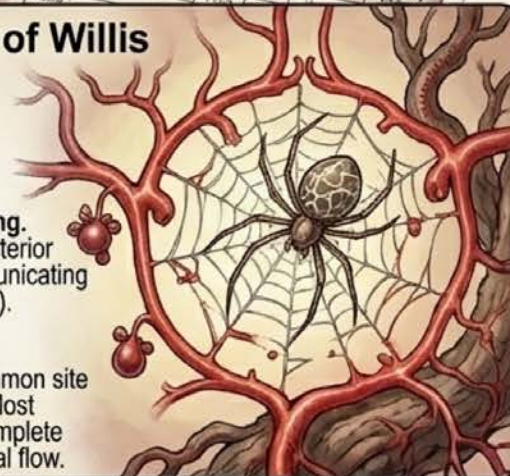
STAT Block
~20% Complete

POWER

The Redundancy Ring.
Connects anterior/posterior
circulations via communicating
arteries (ACoA, PCoA).

LORE

ACoA is the most common site of
berry aneurysms. Most
humans have an incomplete
circle, limiting collateral flow.



Frank the Thrombolysis Window

(Acute Algorithm)

STAT Block
1.9M Neurons/Min

POWER

Time is Brain.
Call EMS -> CT Head ->
Ischemic & $< 4.5h$ = IV alteplase.
If LVO = thrombectomy up to 24h.

LORE

Do NOT lower BP in acute
ischemic stroke unless
 $> 220/120$ or thrombolysis given.



The Stroke Prevention Squad

(Secondary Prevention)

STAT Block
Target $< 130/80$

POWER

The Secondary Shield.
Antiplatelet (non-cardioembolic)
vs. Anticoagulant DOAC
(cardioembolic AF).

LORE

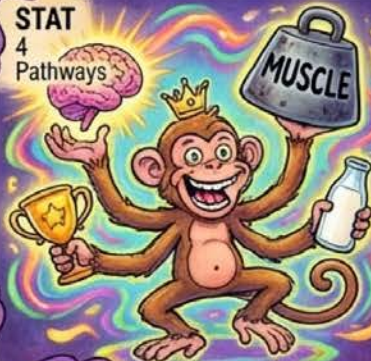
High-intensity statin for ALL ischemic
strokes regardless of baseline LDL.
Carotid endarterectomy for symptomatic stenosis $> 50\%$.



KayfaBIZARRO!

STAT

4 Pathways



Dopa the Dopamine

(Pathways)

POWER

The Quad-Core Modulator.
Nigrostriatal (movement),
Mesolimbic (reward),
Mesocortical (motivation),
Tuberoinfundibular (prolactin).

LORE

Block D2 for psychosis, but you will worsen movement and raise prolactin simultaneously.

STAT

80% Loss



Perry Parkinson

(Parkinson's Disease)

POWER

The Lewy Body Eraser.
Resting tremor (4-6 Hz),
lead-pipe rigidity, and
mandatory bradykinesia.

LORE

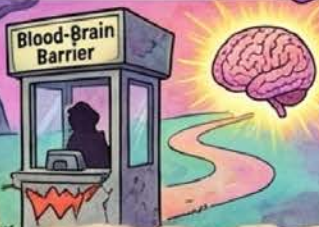
Loss of substantia nigra pars compacta. Hyposmia and REM sleep behavior disorder predate motor signs by YEARS.

Levo the Levodopa

(Pharmacology)

STAT

1:1 CNS Penetration



POWER: The Trojan Precursor.

Crosses BBB, decarboxylates to dopamine. Carbidopa stops peripheral conversion, saving the stomach.

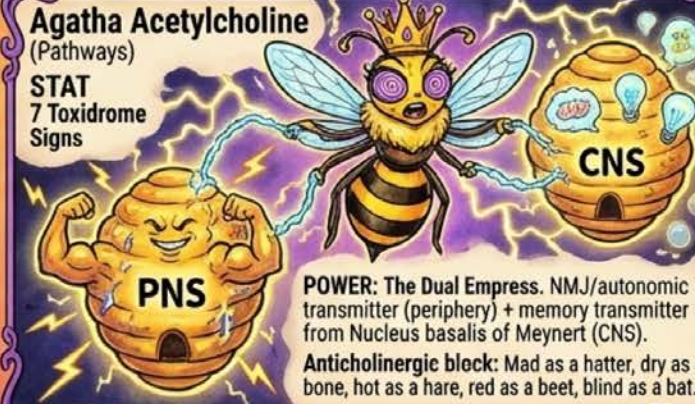
Long-term use causes wearing-off, on-off fluctuations, and pulsatile-induced dyskinesias.

Agatha Acetylcholine

(Pathways)

STAT

7 Toxidrome Signs



POWER: The Dual Empress. NMJ/autonomic transmitter (periphery) + memory transmitter from Nucleus basalis of Meynert (CNS).

Anticholinergic block: Mad as a hatter, dry as a bone, hot as a hare, red as a beet, blind as a bat.



Alex Alzheimer

(Alzheimer's Disease)

STAT Block: Amyloid + Tau

POWER:

The Slow Eraser

Episodic memory loss first -> language/visuospatial -> global decline.

LORE:

Cholinesterase inhibitors (donepezil) increase tone but do not halt progression. Anti-amyloids are the new frontier.



Gabby GABA

(Inhibitory Transmitter)

STAT Block: Chloride Influx

POWER:

The Brain's Brake

Opens ligand-gated chloride channels to hyperpolarize the neuron.

LORE:

Benzodiazepines increase opening frequency. Barbiturates increase duration (and can kill in overdose by acting as direct agonists).



Greta Glutamate

(Excitatory Transmitter)

STAT Block: Mg^{2+} Block Removed

POWER:

The Excitotoxic Accelerator

Fast excitatory transmission via NMDA/AMPA. NMDA requires coincidence detection.

LORE:

In ischemia, massive glutamate release causes unchecked calcium influx and cell death (excitotoxicity).



Serotonin Sally

(5-HT Pathways)

STAT Block: 3-Part Triad

POWER:

The Raphe Projector

Regulates mood, sleep, appetite. Blocked by SSRIs.

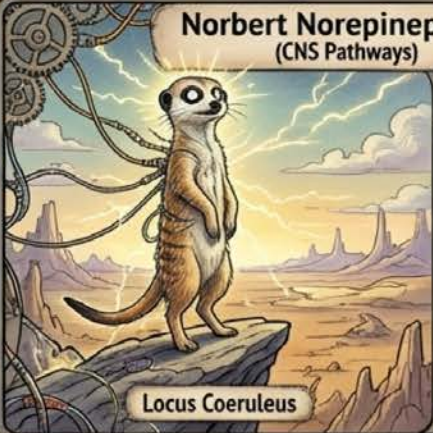
LORE:

Serotonin Syndrome

= Mental status change
+ Autonomic instability
+ Neuromuscular clonus/hyperreflexia (differentiates from NMS rigidity).



KayfaBIZARRO!



Norbert Norepinephrine
(CNS Pathways)

SERT + NET Blocked

POWER
The Sympathetic Driver.
Arousal, attention, stress.
SNRIs block both serotonin and norepinephrine reuptake.

LORE
SNRIs (venlafaxine/duloxetine) are highly effective for both depression AND neuropathic pain via descending pain pathways.

Locus Coeruleus





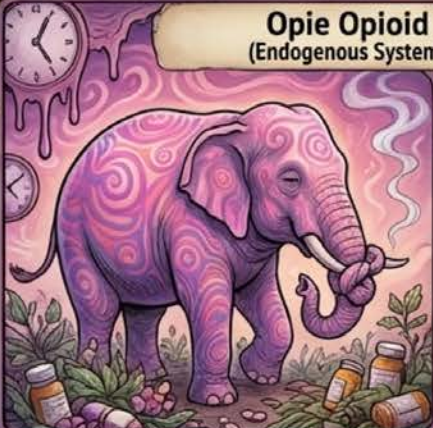
Histo the Histamine
(Wakefulness)

2 Receptor Targets

POWER
The Wakefulness Switch.
H1 drives arousal in CNS.
Loss of orexin activation causes narcolepsy.

LORE
First-generation antihistamines cross the BBB causing profound sedation. Second-generation do not.

Hypothalamus



Opie Opioid
(Endogenous System)

Triad of Overdose

POWER:
The Endogenous Suppressor.
Mu receptor activation gives analgesia, euphoria, miosis, and respiratory depression.

LORE:
Overdose triad = miosis, resp depression, reduced consciousness.
Reverse with Naloxone (redose required for fentanyl).





Endo the Endocannabinoid
(Retrograde Messenger)

Retrograde

POWER:
The Backwards Messenger.
Released postsynaptically, travels backwards to inhibit presynaptic neurotransmitter release via CB1.

LORE:
A unique retrograde signaling mechanism. Epidiolex (CBD) approved for severe childhood epilepsies.

KayfaBOGGLE!

Addie Adenosine (Neuromodulator)

STAT Block: A1/A2A Receptors

POWER:

The Sleep Accumulator

Builds up during wakefulness to apply brakes.

Caffeine is a non-selective antagonist.

LORE:

Exogenous IV bolus briefly blocks AV node conduction—used therapeutically to terminate SVT.

Nicky Nitric Oxide (Gaseous Transmitter)

STAT Block: 0 Receptors

POWER:

The Receptorless Gas

Synthesized on demand.

Diffuses freely across membranes to activate cGMP directly.

LORE:

The primary vascular vasodilator. Sildenafil blocks phosphodiesterase, potentiating this pathway. Toxic in stroke.

Eddie Epilepsy (Classification)

STAT Block: ≥ 2 Unprovoked

POWER:

The ILAE Classifier

Focal (aware/impaired) vs. Generalized.

LORE:

A single seizure is NOT epilepsy unless there is a high recurrence risk (e.g., epileptiform EEG or structural lesion).

Abby Absence (Childhood Absence)

STAT Block: 3 Hz Spike & Wave

POWER:

The Brief Interrupter

5-30 second staring spells, dozens of times daily. Abrupt onset/offset, NO post-ictal confusion.

LORE:

Driven by T-type calcium channels. Ethosuximide is first-line over valproate for girls (avoids teratogenicity).

3Hz Spike-and-Wave

FDA WARNING
THIS WARNING RECALDES
TERATOGENIC RISK
FETAL HARM
OF TEARING LABEL
CONTRAINDICATED IN PREGNANCY
For contraindication be interpreted on
pharmacological basis on drug

WARNING
THIS WARNING RECALDES
TERATOGENIC RISK
FETAL HARM
OF TEARING LABEL
CONTRAINDICATED IN PREGNANCY
For contraindication be interpreted on
pharmacological basis on drug

KayfaEMERGENCY!

Tonic Tom (GTCS)



STAT Block
2 Phases

POWER

The Grand Mal
Tonic phase (sustained contraction, apnea) ->
Clonic phase (rhythmic jerking).

LORE

Always followed by post-ictal confusion/fatigue.
Distinguish from syncope by prolonged post-ictal state.

Stan Status (Status Epilepticus)



STAT Block
> 5 Minutes

POWER

The Time-Critical Emergency
>5 min continuous, or
2+ seizures without recovery in between.

LORE

5-16m: IV Lorazepam
>20m: Levetiracetam/ Valproate IV.
Prolonged seizures internalize GABA receptors, becoming refractory.

What the FLUFF?!

What the FLUFF?!

Valerie Valproate (Broad-spectrum)



STAT Block
0% in Pregnancy

POWER

The Broad-Spectrum Weapon
Blocks Na⁺, blocks T-type Ca²⁺, enhances GABA.
Works for focal, generalized, bipolar, migraine.

LORE

Severe neural tube defects. Absolutely avoid in women of childbearing potential unless NO alternative exists.

Liam Levetiracetam (Keppra)



STAT Block
SV2A Target

POWER

The Modern Workhorse
Binds SV2A modulating neurotransmitter release.
Minimal drug interactions.

LORE

The preferred broad-spectrum agent in women of childbearing age. Counsel patients on mood disturbance/irritability.

TERATOGENIC
INGINITY BY
FETAL HARM
CONTRAINDICATED IN PREGNANCY

KayfaBIZARRO!

Lammy Lamotrigine (Sodium channel blocker)

STAT Block
2x Level w/ Valproate



POWER

The Slow Titrator
Blocks voltage-gated sodium channels. Must be titrated extremely slowly to avoid Stevens-Johnson Syndrome.

LORE

Valproate **DOUBLES** Lamotrigine levels. Carbamazepine halves them. Adjust dose carefully.

Carla Carbamazepine (Enzyme inducer)

STAT Block
CYP3A4 Inducer



POWER

The Auto-Inducer
First-line focal epilepsy and trigeminal neuralgia. Induces its own metabolism.

LORE

Devastating drug-drug interactions. Mandatory HLA-B*1502 screening for SJS risk in East Asian ancestry.

Manny Meningitis (Bacterial Meningitis)

STAT Block
3-Sign Triad



POWER

The Ceftriaxone Race.
Fever, neck stiffness, photophobia. Kernig's + Brudzinski's positive.

LORE

Give IV Ceftriaxone immediately. Do NOT delay antibiotics for CT or LP. Add dexamethasone for pneumococcal.

Herbie Herpes Encephalitis (HSE)

STAT Block
14-21 Days
Aciclovir



POWER

The Temporal Invader.
Fever, headache, severe behavioral change, olfactory hallucinations.

LORE

T2/FLAIR hyperintensity in temporal lobes. Start IV Aciclovir empirically BEFORE PCR results return. Delay is fatal.

Spinal Fluid

KayfaBIZARRO!

KayfaBIZARRO!

KayfaNEURO!

Stay fluffy!



Guillain the Ascending (GBS)

STAT Block
FVC < 15-20 ml/kg

Slowly
Frozen



POWER

The Post-Infectious Climb. Acute immune polyneuropathy. Ascending areflexic weakness.

LORE

Albuminocytological dissociation in CSF. Monitor FVC closely; intubate early. Steroids **DO NOT WORK**; use IVIg or Plasma Exchange.

The Antibody Attacker (Myasthenia Gravis)

STAT Block
80-85% Anti-AChR



POWER

The Fatigable Drop. Autoantibodies destroy postsynaptic nicotinic receptors. Fatigable weakness (worse with repetition).

LORE

Myasthenic crisis = bulbar/respiratory failure. Differentiate from cholinergic crisis via Tensilon test. Treat with Pyridostigmine.

Migs the Migraine (CSD Mechanism)

STAT Block
4-72 Hours



POWER

Cortical Spreading Depression. Unilateral, throbbing, nausea, photophobia. 30% have aura.

LORE

Triptans (5-HT agonists) cause vasoconstriction—absolutely contraindicated in hemiplegic or brainstem aura migraines.

Hugh Hydrocephalus (Obstructive vs. Communicating)

STAT Block
Hakim's Triad



POWER

The Fluid Trap. Obstructive (aqueduct stenosis) vs. Communicating (impaired reabsorption).

LORE

NPH Triad: Gait apraxia, urinary incontinence, dementia. LP drainage test predicts shunt response.

Ida Intracranial Pressure

(Raised ICP)



STAT Block: Target 5-15 mmHg

POWER:

Cushing's Reflex.
Hypertension + bradycardia + irregular respiration = pre-herniation.

LORE:

Head 30 dsg up, mannitol.
Hyperventilation is strictly temporary (causes vasoconstriction to buy time).
Steroids ONLY for tumor/abscess.

BLÖDSINN!



Terry TBI

(Traumatic Brain Injury)



STAT Block: CPP 60-70 mmHg

POWER:

The Secondary Preventer.
Primary impact is done, your job is stopping secondary ischemia/hypoxia.

LORE:

Cerebral Perfusion Pressure (CPP) = MAP - ICP.
Hypotension starves the brain; hypertension worsens edema.

BLÖDSINN!



Molly MS

(Multiple Sclerosis)



STAT Block: 85% RRMS

POWER:

Dissemination in Time & Space.
T-cell autoimmune demyelination.
Optic neuritis, INO, Lhermitte's phenomenon.

LORE:

Uhthoff's phenomenon: Heat worsens symptoms due to conduction failure in already demyelinated axons.

BLÖDSINN!



Hugo Huntington

(Huntington's Disease)



STAT Block: CAG Repeat > 36

POWER:

The Autosomal Expander.
Trinucleotide repeat expansion.
Anticipation: worsens in successive generations.

LORE:

Caudate nucleus degeneration disinhibits the motor pathway
→ chorea, cognitive decline, psychiatric features.



Alan ALS (Motor Neuron Disease)



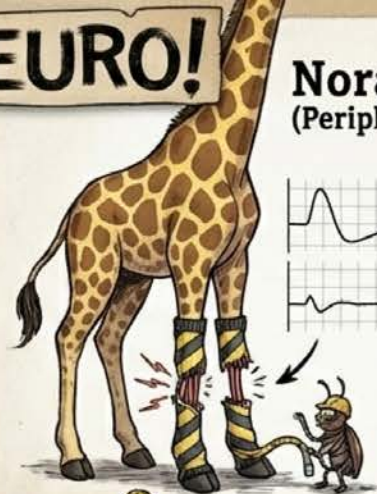
STAT Block: UMN + LMN

POWER:
The Dual Tract Destroyer.
Attacks corticospinal tract AND
anterior horn cells.

LORE:
Co-existence of UMN and LMN signs
in the same territory is the clinical
hallmark.
Riluzole extends survival modestly.

KayfaNEURO!

Nora Neuropathy (Peripheral Neuropathy)



STAT Block: 2 NCS Patterns

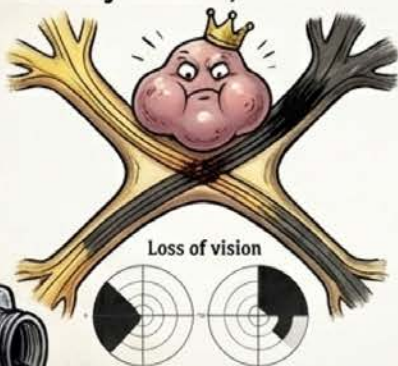
POWER:
The Pattern Mapper.
Length-dependent (stocking-glove),
mononeuritis multiplex, or radiculopathy.

LORE:
Nerve conduction distinguishes
Demyelinating (slow velocity) vs.
Axonal (reduced amplitude).
Diabetes is the commonest cause.

KayfaBONGO!



Ronnie Raised Prolactin (Pituitary Tumors)



STAT Block: Bitemporal Hemianopia

POWER:
The Chiasm Compressor.
Most common pituitary tumor.
Causes galactorrhea, amenorrhea,
and visual field loss.

LORE:
Compresses the crossing nasal fibers
of the optic chiasm, causing loss of
peripheral vision on both sides.

Bella Bell's Palsy (LMN VII Palsy)



STAT Block: NNT ~10 (Steroids)

POWER:
The LMN VII Flattening.
Unilateral LMN facial palsy. Affects
the ENTIRE face, distinguishing
it from an UMN stroke.

LORE:
UMN (stroke) spares the forehead
due to bilateral cortical innervation.
Give prednisolone <72h.
Tape the eye shut!

Terry Trigeminal

(Trigeminal Neuralgia)

STAT: YOU!



STAT Block: NNT -3 (Carbamazepine)

POWER:
The Vascular Compressor.
Lancinating, electric-shock pain in
V2/V3 distribution triggered by
light touch.

LORE:
Caused by vascular compression
of the nerve root.
Carbamazepine is highly effective
first-line.



The Neurology Attending

(Master Review Card)



STAT Block: 3 Simultaneous Patients

POWER:
Master Synthesis. A 70yo with
aphasia/arm weakness.
A 25yo with ascending areflexia.
A 45yo with visual loss and headache.

LORE:
Localize the lesion. Name the
syndrome. Select the treatment.
Map every decision. There is no
hiding on the ward.

Ida IHH

(Idiopathic Intracranial Hypertension)



STAT Block: Opening Pressure > 25

POWER:
The Pseudotumor.
Raised ICP without a mass.
Daily headache, pulsatile tinnitus,
visual obscurations.

LORE:
Untreated leads to blindness.
Weight loss is the definitive treatment;
Acetazolamide reduces CSF secretion.

Uncle FrizzleBob's Blank Sketch Card

(DIY Template)



STAT Block: STAT: YOU

POWER:
Draw Your Own.
The ultimate Kayfabizzaro challenge.

LORE:
The mechanics are real. The lore
is real. Only the animals are fake.
Draw your next patient. Stay fluffy!